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571-272-7822

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

JSR CORPORATION and JSR LIFE SCIENCES, LLC,
Petitioner,

v.

CYTIVA BIOPROCESS R&D AB,
Patent Owner.

IPR2022-00042
IPR2022-00045
Patent 10,875,007 B2

Before ULRIKE W. JENKS, SHERIDAN K. SNEDDEN, and SUSAN L. C. MITCHELL, *Administrative Patent Judges.*

SNEDDEN, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision
Determining Some Challenged Claims Unpatentable
35 U.S.C. § 318

I. INTRODUCTION

This is a Final Written Decision in an *inter partes* review of claims 1–14, 16–32, and 34–37 ("the challenged claims) of U.S. Patent No. 10,875,007 B2 (Ex. 1001, "the '007 patent"). We have jurisdiction under 35 U.S.C. § 6, and enter this Final Written Decision pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73. For the reasons set forth below, we determine that JSR Corporation and JSR Life Sciences, LLC (collectively, "Petitioner") has shown, by a preponderance of the evidence, that some of the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e).

A. *Consolidated Proceedings*

The two captioned proceedings (IPR2022-00042 and IPR2022-00045 (or "the '045 IPR")) involve the '007 patent and challenge the same set of claims. The asserted grounds and prior art contentions are different in each proceeding. Consolidation is appropriate where, as here, the Board can more efficiently handle the common issues and evidence, and also remain consistent across proceedings. Under 35 U.S.C. § 315(d), the Director may determine the manner in which these pending proceedings may proceed, including "providing for stay, transfer, consolidation, or termination of any such matter or proceeding." *See also* 37 C.F.R. § 42.4(a) ("The Board institutes the trial on behalf of the Director."). There is no specific Board rule that governs consolidation of cases. But 37 C.F.R. § 42.5(a) allows the Board to determine a proper course of conduct in a proceeding for any situation not specifically covered by the rules and to enter non-final orders to administer the proceeding. Therefore, on behalf of the Director under § 315(d), and for a more efficient administration of these proceedings, we consolidate IPR2022-00042 and IPR2022-00045 for purposes of rendering this Final Written Decision.

B. *Evidence*

Petitioner relies upon information that includes the following:

Ex. 1004, M. Linhult, *et al.*, *Improving the Tolerance of a Protein A Analogue to Repeated Alkaline Exposures Using a Bypass Mutagenesis Approach*, 55 PROTEINS: STRUCTURE, FUNCTION, AND BIOINF., 407–16 (2004) ("Linhult").

Ex. 1005, L. Abrahmsén, et al., U.S. Patent No. 5,143,844 (issued Sept. 1, 1992) ("Abrahmsén").

Ex. 1006, S. Hober, PCT Publication No. WO 03/080655 A1 (published Oct. 2, 2003) ("Hober").

Ex. 1018, H. Berg., U.S. Patent Application Publication No. 2006/0134805 (published June 22, 2006) ("Berg").

C. *Procedural History*

Petitioner filed a Petition for an *inter partes* review of the challenged claims under 35 U.S.C. § 311. Paper 1[1] ("Pet."). Petitioner supported the Petition with the Declaration of Dr. Steven M. Cramer. Ex. 1002. Cytiva Bioprocess R&D AB ("Patent Owner") filed a Patent Owner Preliminary Response to the Petition. Paper 8.

On May 19, 2022, pursuant to 35 U.S.C. § 314(a), we instituted trial ("Decision" or "Dec." (Paper 10)) to determine whether any challenged claim of the '007 patent is unpatentable.

[1] We note that the evidence filed in both proceedings is generally consistent in having the same exhibit number. Therefore, we reference exhibits and paper numbers as they appear in the record of IPR2022-00042, unless otherwise noted.

In IPR2022-00042, Petitioner asserts the following grounds of unpatentability (Pet. 4):

Claim(s) Challenged	35 U.S.C. §[2]	Reference(s)/Basis
1–11, 20–29	103(a)	Linhult, Abrahmsén
1–14, 16–32, 34–37	103(a)	Linhult, Hober
1–14, 16–32, 34–37	103(a)	Linhult, Abrahmsén, Hober
1–14, 16–32, 34–37	103(a)	Abrahmsén, Hober

In IPR2022-00045, Petitioner asserts the following grounds of unpatentability ('045 IPR Pet. 4):

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1–14, 16–18, 20–32, 34–36	103(a)	Berg, Linhult
4–8, 10, 19, 22–26, 28, 37	103(a)	Berg, Linhult, Hober
1–3, 9, 10, 12–14, 16–18, 20, 21, 27, 28, 30–32, 34–36	103(a)	Berg, Abrahmsén
4–8, 10, 11, 19, 22, 26, 28, 29, 37	103(a)	Berg, Abrahmsén, Hober

Patent Owner filed a Patent Owner Response to the Petition. Paper 15 ("PO Resp."). Patent Owner supported the Response with the Declaration of

[2] The Leahy-Smith America Invents Act ("AIA") included revisions to 35 U.S.C. § 103 that became effective on March 16, 2013. Because the '007 patent issued from an application claims priority from an application filed before March 16, 2013, we apply the pre-AIA versions of the statutory bases for unpatentability.

Dr. Daniel Bracewell (Ex. 2025). *See* PO Resp., iv (Exhibit List). Petitioner filed a Reply to the Patent Owner Response. Paper 28 ("Reply"). Petitioner supported the Reply with a Reply Declaration from Dr. Steven M. Cramer. Ex. 1061. Patent Owner filed a Sur-reply to Petitioner's Reply. Paper 34 ("Sur-reply").

On February 16, 2023, the parties presented arguments at an oral hearing. Paper 35. The hearing transcript has been entered in the record. Paper 39 ("Tr.").

For the reasons set forth below, we determine that Petitioner has shown by a preponderance of the evidence that claims 1–10, 12–14, 16–28, 30–32, and 34–37 of the '007 patent are unpatentable, but find that Petitioner has not shown by a preponderance of the evidence that claims 11 and 29 are unpatentable.

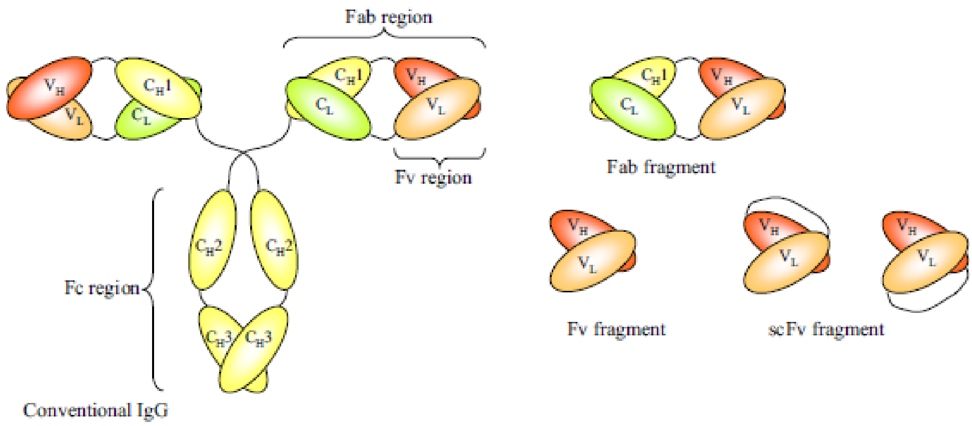
D. *Related Matters*

The '007 patent is at issue in *Cytiva Bioprocess R&D et al. v. JSR Corp. et al.*, Case No. 21-310-RGA (D. Del.). Pet. 2; Paper 5, 1.

In addition to the '007 patent, Petitioner filed Petitions for *inter partes* review of related U.S. patents as follows: U.S. Patent No. 10,343,142 B2 ("the '142 patent") in IPR2022-00041 and IPR2022-00044; and U.S. Patent No. 10,213,765 B2 ("the '765 patent") in IPR2022-00036 and IPR2022-00043. Pet. 2–3; Paper 5, 1–2. The '007 patent is a continuation of the '142 patent which is a continuation of the '765 patent. Ex. 1001, code (60).

E. *Subject matter background*

Antibodies (also called immunoglobulins) are glycoproteins, which specifically recognize foreign molecules. These recognized foreign molecules are called antigens. Ex. 2001, 1. A schematic representation of the structure of a conventional IgG and fragments is shown below:

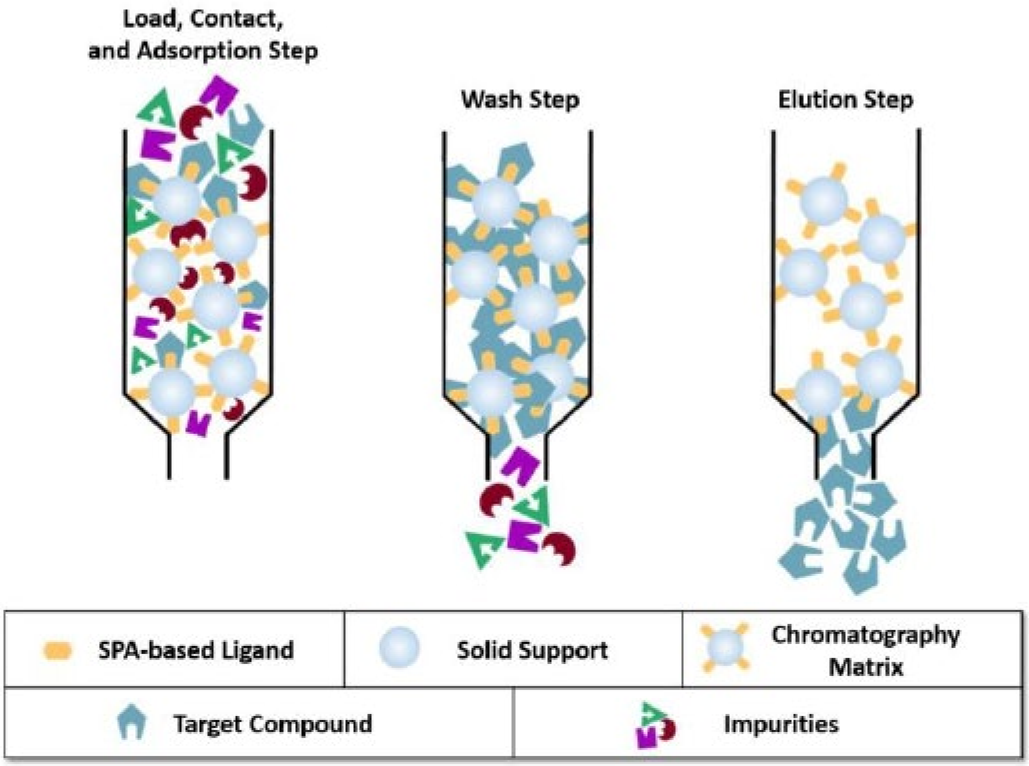


The figure (Ex. 2001, 2 (Fig. 1)), reproduced above, shows

> the structure of a conventional IgG and fragments that can be generated thereof. The constant heavy-chain domains C_H1, C_H2 and C_H3 are shown in yellow, the constant light-chain domain (C_L) in green and the variable heavy-chain (V_H) or light-chain (V_L) domains in red and orange, respectively. The antigen binding domains of a conventional antibody are Fabs and Fv fragments. Fab fragments can be generated by papain digestion. Fvs are the smallest fragments with an intact antigen-binding domain. They can be generated by enzymatic approaches or expression of the relevant gene fragments (the recombinant version). In the recombinant single-chain Fv fragment, the variable domains are joined by a peptide linker. Both possible configurations of the variable domains are shown, i.e. the carboxyl terminus of VH fused to the N-terminus of VL and vice versa.

Ex. 2001, 2; *see also* PO Resp. 5.

Below is a generic, exemplary schematic that shows how affinity purification typically works:



The figure shows the schematic of the loading, contact, and adsorbing step onto a column, followed by the wash step, and finally the elution and collection of the target compound. Ex. 1002 ¶ 24 (citing Ex. 1014 §§ 1.1, 4.2.); *see also* PO Resp. 7 ("In a typical process, the composition containing the desired antibody then is loaded onto (i.e., pumped or injected into) the column."); Pet. 6; *see generally* Ex. 1014.

F. *The '007 patent (Ex. 1001)*

The '007 patent discloses an affinity ligand useful for isolating antibodies or antibody fragments. *See* Ex. 1001, Abstr., 1:43–49. Affinity ligands were previously used to capture antibodies (immunoglobulin proteins) in chromatography matrices. *See id.* at 2:1–19. After each use, chromatography matrices are cleaned using an alkaline protocol "known as Cleaning In Place (CIP)." *See id.* at 2:20–54. CIP damages protein-based affinity ligands through deamidation and cleavage of peptides. *Id.* at 2:35–54.

The ’007 discloses prior efforts to genetically engineer protein-based affinity ligands to improve stability to alkaline cleaning, e.g., CIP. *See id.* at 2:53–54; 3:3–55. Specifically, the ’007 patent describes a modified affinity ligand based on *Staphylococcus* protein A (“SpA”). SpA was “widely used” as an affinity chromatography ligand due to its ability to bind to antibodies without affecting the antibodies’ ability to bind to antigens. *Id.* at 2:55–68. However, the ’007 patent discloses that unmodified SpA required milder cleaning conditions than conventional CIP to prevent damaging the SpA ligand. *Id.* at 3:18–25. Accordingly, the ’007 patent describes a need for modifying SpA to improve alkaline resistance while maintaining binding selectivity. *Id.* at 3:25–28.

The ’007 patent describes prior efforts to modify SpA through its constituent domains. *See id.* at 3:42–55. SpA is composed of five domains, designated as domains E, D, A, B, and C, which are able to bind to antibodies at the antibodies’ fragment crystallizable (“Fc”) region. *Id.* at 2:60–65. The ’007 patent describes a known modified SpA B-domain with “increased chemical stability at pH-values of up to about 13–14.” *Id.* at 3:42–55 (citing WO 03/080655). The increased stability results from mutating “at least one asparagine residue . . . to an amino acid other than glutamine or aspartic acid,” as it was known that asparagine and glutamine residues were sensitive to deamidation and cleavage in alkaline conditions. *See id.* at 2:42–47; 3:42–55.

Against this background, the ’007 patent discloses an affinity ligand based on SpA domain C that “is capable of withstanding repeated cleaning-in-place cycles." *Id.* at 4:9–13. The Specification discloses that "Domain C ligand, which contains as many as six asparagine residues, was not . . . expected to present any substantial alkaline-stability as compared to protein A." *Id.* at 5:54–56. The asparagine residues are shown in the amino acid sequence of wild-type SpA Domain C SEQ ID NO 1. *Id.* at 6:40–43; 15:1–25. The Specification describes "a specific embodiment [of] the chromatography ligand according to the invention [that] comprises SpA Domain C, as shown in SEQ ID NO 1, which in addition comprises the mutation G29A." *Id.* at 6:56–61. In other words, the modified SpA Domain C includes alanine (A) instead of glycine (G) at position 29 of the amino acid sequence. *See id.* at 15:52–54. The '007 patent further describes a multimeric chromatography ligand including at least two Domain C units, or functional variants thereof. *Id.* at 7:35–38.

1. Illustrative Claims

Claims 1 and 20 are the independent claims, reproduced below, and are illustrative of the claimed subject matter of the '007 patent.

> 1. A process for isolating one or more target compound(s), the process comprising:
>
> (a) contacting a first liquid with a chromatography matrix, the first liquid comprising the target compound(s) and the chromatography matrix comprising:
>
> (i) a solid support; and
>
> (ii) at least one ligand coupled to the solid support, the ligand capable of binding the one or more target compound(s) and comprising at least two polypeptides, wherein the amino acid sequence of each polypeptide comprises at least 52 contiguous amino acids of a modified SEQ ID NO. 1, and wherein the modified SEQ ID NO. 1 has an alanine (A) instead of glycine (G)

at a position corresponding to position 29 of SEQ ID NO. 1; and

(b) adsorbing the target compound(s) to the ligand;

(c) eluting the compound(s) by passing a second liquid through the chromatography matrix that releases the compound(s) from the ligand; and,

(d) performing a cleaning in place (CIP) process involving exposing the chromatography matrix to a CIP solution with a NaOH concentration of at least 0.1 M.

20. A process for isolating one or more target compound(s), the process comprising:

(a) contacting a first liquid with a chromatography matrix, the first liquid comprising the target compound(s) and the chromatography matrix comprising:

(i) a solid support; and

(ii) at least one ligand coupled to the solid support, the ligand capable of binding the one or more target compound(s) and comprising at least two polypeptides, wherein the amino acid sequence of each polypeptide comprises at least 55 amino acids in alignment with SEQ ID NO. 1, and wherein each polypeptide has an alanine (A) instead of glycine (G) at a position corresponding to position 29 of SEQ ID NO. 1;

(b) adsorbing the target compound(s) to the ligand; and,

(d) performing a clean in place (CIP) process involving exposing the chromatography matrix to a CIP solution with a NaOH concentration of at least 0.1 M.

Ex. 1001, 15:40–61, 17:22–40

II. ANALYSIS

A. *Principles of Law*

"In an IPR, the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to Patent Owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) (discussing the burden of proof in *inter partes* review).

Petitioner must demonstrate by a preponderance of the evidence[3] that the claims are unpatentable. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d). A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time of the invention to a person having ordinary skill in the art. *KSR Int'l Co. v. Teleflex, Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including "the scope and content of the prior art"; "differences between the prior art and the claims at issue"; "the level of ordinary skill in the art;" and "objective evidence of non-obviousness." *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

[3] The burden of showing something by a preponderance of the evidence requires the trier of fact to believe that the existence of a fact is more probable than its nonexistence before the trier of fact may find in favor of the party who carries the burden. *Concrete Pipe & Prods. of Cal., Inc. v. Constr. Laborers Pension Tr. for S. Cal.*, 508 U.S. 602, 622 (1993).

In analyzing the obviousness of a combination of prior art elements, it can be important to identify a reason that would have prompted one of skill in the art "to combine . . . known elements in the fashion claimed by the patent at issue." *KSR*, 550 U.S. at 418. A precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *Id.* Rather, "any need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed." *Id.* at 420. Accordingly, a party that petitions the Board for a determination of unpatentability based on obviousness must show that "a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so." *In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (internal quotations and citations omitted).

B. *Level of Ordinary Skill in the Art*

In determining the level of skill in the art, we consider the type of problems encountered in the art, the prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus. Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986); *Orthopedic Equip. Co. v. United States*, 702 F.2d 1005, 1011 (Fed. Cir. 1983).

Petitioner asserts that a person of ordinary skill in the art would have had

> (1) at least an advanced degree (*e.g.*, a Master's or Ph.D.) in biochemistry, process chemistry, protein chemistry, chemical engineering, molecular and structural biology, biochemical engineering, or similar disciplines; (2) several years of post-graduate training or related experience (including industry experience) in one or more of these areas; and (3) an understanding of the various factors involved in purifying proteins using chromatography.[] Such a person would have had multiple years of experience with affinity ligand design and protein purification.

Pet. 9–10 (citing Ex. 1002 ¶¶ 13–14). Patent Owner does not dispute Petitioner's definition of the person of ordinary skill. *See generally* PO Resp. Because Petitioner's proposed definition is unopposed and appears consistent with the Specification and art of record, we apply it here.

C. *Claim Construction*

The Board applies the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 42.200(b) (2021). Under that standard, claim terms "are generally given their ordinary and customary meaning" as understood by a person of ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Claims 1 and 20 recite "at least one ligand coupled to the solid support, the ligand capable of binding the one or more target compound(s) and comprising at least two polypeptides." Ex. 1001, 15:46–49; 17:28–31. Petitioner argues that "'the ligand comprising at least two polypeptides' refers to a multimeric ligand (such as a tetramer) comprised of multiple polypeptides, each of which is a monomer." Pet. 19–20 (citing Ex. 1002 ¶¶ 43–45). Petitioner argues that this construction is consistent with Patent

Owner's implicit construction in the Delaware litigation between the parties. *Id.*

Patent Owner does not challenge Petitioner's claim construction. *See generally* PO Resp.

According to the Specification, "the present invention . . . relates to a multimeric chromatography ligand (also denoted a 'multimer') comprised of at least two domain C units, or a functional fragments [sic] or variants thereof." Ex. 1001, 7:35–38. The Specification additionally recites that a multimer containing only domain C units can, however, include linkers. *Id.* at 7:39–41. In addition, the Specification describes that "the multimer comprises one or more additional units, which are different from Domain C." *Id.* at 7:52–53. Based on these disclosures in the Specification, a multimer is composed of at least two or more monomers. Because Petitioner's construction is consistent with the '007 patent's express construction of the term, we apply that construction for the purposes of this decision.

Having considered the parties' positions and evidence of record, we determine that the express construction of any other claim term is unnecessary to resolve the disputed issues in this matter. *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) ("[W]e need only construe terms 'that are in controversy, and only to the extent necessary to resolve the controversy.'" (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.,* 200 F.3d 795, 803 (Fed. Cir. 1999))). To the extent further discussion of the meaning of any claim term is necessary to our decision, we provide that discussion below in our analysis of the asserted grounds of unpatentability.

D. *Overview of Asserted References*

1. Linhult (Ex. 1004)

Linhult is titled "Improving the Tolerance of a Protein A Analogue to Repeated Alkaline Exposures Using a Bypass Mutagenesis Approach." Ex. 1004, 1. Linhult discloses that due to the high affinity and selectivity of Staphylococcal protein A (SPA), "it has a widespread use as an affinity ligand for capture and purification of antibodies" but that "it is desirable to further improve the stability in order to enable an SPA-based affinity medium to withstand even longer exposure to the harsh conditions associated with cleaning-in-place (CIP) procedures." *Id.*, Abstr. Linhult discloses, "[t]o further increase the alkaline tolerance of SPA, we chose to work with Z, which is a small protein derived from the B domain of SPA." *Id.* at 2.

Figures 1A and 1B of Linhult are reproduced below.

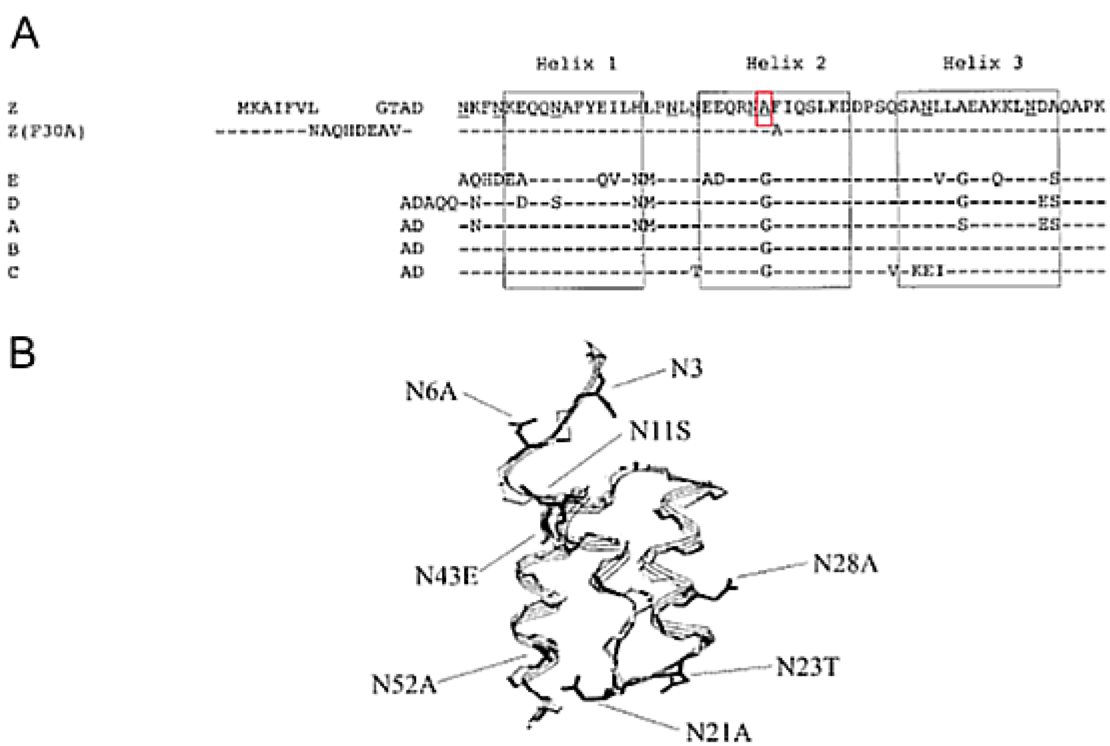


Figure 1A shows "[a]mino acid alignments of the Z, Z(F30A) and the five homologous domains (E, D, A, B, and C)" in which the horizontal lines indicate amino acid identity and "one glycine in the B domain [is] replaced [and] underlined" as annotated by the Board with a red box. *Id.* "Z(F30A), and all mutants thereof includes the same N-terminal as Z(F30A)" and "Z(N23T) was constructed with the same N-terminal as Z." *Id.*[4] Figure 1B shows "[t]he three-dimensional structure of the Z domain" and "the different substitutions are indicated." *Id.* Specifically, Linhult discloses,

> [t]he B domain has been mutated in order to achieve a purification domain resistant to cleavage by hydroxylamine. An exchange of glycine 29 for an alanine has been made in order to avoid the amino acid combination asparagine–glycine, which is a cleavage site for hydroxylamine.[] Asparagine with a succeeding glycine has also been found to be the most sensitive amino acid sequence to alkaline conditions.[] Protein Z is well characterized and extensively used as both ligand and fusion partner in a variety of affinity chromatography systems.

Id. Using a 0.5 M NaOH cleaning agent and "a total exposure time of 7.5 h for Z(F30A) and mutants thereof," Linhult determines that "N23 seems to be very important for the functional stability after alkaline treatment of Z(F30A)" and "Z(F30A, N23T) shows only a 28% decrease in capacity despite the destabilizing F30A-mutation." *Id.* at 410–11; Figs. 2, 3. Linhult reports that "[h]ence, the Z(F30A ,N23T) is almost as tolerant as Z and is thereby the most improved variant with Z(F30A) as scaffold." *Id.* at 411; Figs. 2, 3.

[4] The mutation N23T having a change in amino acid correlates with the amino acid N next to the "Helix 2" box of Figure 1A as annotated by Petitioner. *See* Pet. 12.

Linhult further discloses that "Z, Z(F30A), and mutated variants were covalently coupled to HiTrap™ affinity columns," that "[t]he Z domain includes 8 asparagines (N3, N6, N11, N21, N23, N28, N43, and N52; Fig. 1)," and that "since the amino acid is located outside the structured part of the domain, it will most likely be easily replaceable during a multimerization of the domain to achieve a protein A–like molecule." *Id.* at 410. Linhult confirms that "the affinity between Z(F30A) and IgG was retained despite the mutation." *Id.* In Linhult's studies, "[h]uman polyclonal IgG in TST was prepared and injected onto the columns in excess" and "[a] standard affinity chromatography protocol was followed." *Id.*

2. Abrahmsén (Ex. 1005)

Abrahmsén "relates to a recombinant DNA fragment coding for an immunoglobulin G ([]IgG) binding domain related to staphylococcal protein A . . . and to a process for cleavage of a fused protein expressed by using such fragment or sequence." Ex. 1005, 1:8–13. Abrahmsén discloses that "[b]y making a gene fusion to staphylococcal protein A any gene product can be purified as a fusion protein to protein A and can thus be purified in a single step using IgG affinity chromatography." *Id.* at 1:22–26. Abrahmsén explains that Protein A has "5 Asn-Gly in the IgG binding region of protein A" which "makes the second passage through the column irrelevant as the protein A pieces released from the cleavage will not bind to the IgG." *Id.* at 1:58–63. Abrahmsén provides a solution to this problem "by adapting an IgG binding domain so that no Met and optionally no Asn-Gly is present in the sequence." *Id.* at 1:64–67.

Abrahmsén discloses that in a preferred embodiment, "the glycine codon in the Asn-Gly constellation has been replaced by an alanine codon."

Id. at 2:21–23. In one embodiment, Abrahmsén provides "a recombinant DNA sequence comprising at least two Z-fragments" in which "the number of such amalgamated Z-fragments is preferably within the range 2–15, and particularly within the range 2–10." *Id.* at 2:27–31. Abrahmsén discloses that the recombinant DNA fragment can "cod[e] for any of the E D A B C domains of staphylococcal protein A, wherein the glycine codon(s) in the Asn-Gly coding constellation has been replaced by an alanine codon." *Id.* at 2:32–37. According to Abrahmsén, from a simulation of the Gly to Ala amino acid change in the computer, it was "concluded that this change would not interfere with folding to protein A or binding to IgG." *Id.* at 5:13–16.

3. Hober (Ex. 1006)

Hober "relates to . . . a mutant protein that exhibits improved stability compared to the parental molecule" and "also relates to an affinity separation matrix, wherein a mutant protein according to the invention is used as an affinity ligand." Ex. 1006, 1. Hober discloses that removal of contaminants from the separation matrix involves "a procedure known as cleaning-in-place (CIP)" but "[f]or many affinity chromatography matrices containing proteinaceous affinity ligands," the alkaline environment "is a very harsh condition and consequently results in decreased capacities owing to instability of the ligand." *Id.* at 1–2. According to Hober, stability to alkaline conditions can be engineered into a protein. *Id.* at 2. To improve the stability of a Streptococcal albumin-binding domain (ABD) in alkaline environments, it has been reported to involve the role of peptide conformation in the rate and mechanism of deamidation of asparaginyl residues and that "the shortest deamidation half time have been associated with the sequences -asparagine-glycine and -asparagine-serine." *Id.* at 2. Further, from a study of a mutant of ABD that was created, it was concluded that "all four asparagine residues can be replaced without any significant effect on structure and function." *Id.* at 2–3. Hober points out that the staphylococcal protein A (SPA) contains domains capable of binding to the Fc and Fab portions of IgG immunoglobulins from different species and reagents of this protein with their high affinity and selectivity have found a widespread use in the field of biotechnology. *Id.* at 3. Accordingly, "there is a need in this field to obtain protein ligands capable of binding immunoglobulins, especially via the Fc-fragments thereof, which are also tolerant to one or more cleaning procedures using alkaline agents." *Id.* at 4.

In one embodiment of Hober, a multimer "comprises one or more of the E, D, A, B, and C domains of Staphylococcal protein A" in which "asparagine residues located in loop regions have been mutated to more hydrolysis-stable amino acids" for advantageous structural stability reasons wherein "the glycine residue in position 29 of SEQ ID NO[]. 1 has also been mutated, preferably to, an alanine residue." *Id.* at 12. Hober's SEQ ID NO: 1 and is reproduced below.

```
Ala Asp Asn Lys Phe Asn Lys Glu Gln Gln Asn Ala Phe Tyr Glu Ile
1               5                   10                  15

Leu His Leu Pro Asn Leu Asn Glu Glu Gln Arg Asn Gly Phe Ile Gln
            20                  25                  30

Ser Leu Lys Asp Asp Pro Ser Gln Ser Ala Asn Leu Leu Ala Glu Ala
        35                  40                  45

Lys Lys Leu Asn Asp Ala Gln Ala Pro Lys
    50                  55
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Id. at SEQUENCE LISTING 1. SEQ ID NO: 1 shows a domain of Staphylococcus aureus having Glycine (Gly) as an amino acid at the position 29, as annotated by the Board via red highlighting.

Hober further discloses that its matrix for affinity separation "comprises ligands that comprise immunoglobulin-binding protein coupled to a solid support, in which [in the] protein at least one asparagine residue has been mutated to an amino acid other than glutamine." *Id.* at 13. For its method of isolating an immunoglobulin, Hober discloses "in a first step, a solution comprising the target compounds, . . . is passed over a separation matrix under conditions allowing adsorption of the target compound to ligands present on said matrix" and "[i]n a next step, a second solution denoted an eluent is passed over the matrix under conditions that provide desorption, i.e. release of the target compound." *Id.* at 13.

E. *Obviousness in view of Linhult, Abrahmsén, and Hober*

1. Petitioner's Contentions

a) Claims 1, 2, and 20

Petitioner contends that "*Linhult* describes the common use of chromatography matrices in the biotechnology field, and, more specifically, SPA-based chromatography matrices to isolate target compounds." Pet. 16 (citing Ex. 1002 ¶ 83). Petitioner contends that "*Linhult* describes a process whereby a '[h]uman polyclonal IgG in TST[5] was prepared and injected onto the columns in excess,' '[a] standard affinity chromatography protocol was followed,' and 'eluted material was detected.'" Pet. 17 (citing Ex. 1004,

[5] TST is a solution containing 25 mM Tris-HCl pH 7.5, 150 mM NaCl, 1.25 mM EDTA, 0.05% Tween 20. Ex. 1004, 4.

4). Petitioner contends that a person of ordinary skill in the art "would have further understood a 'standard affinity chromatography protocol' would involve the well-known and conventional step of loading a liquid comprising the target compound onto the column, thereby allowing the liquid to contact the recited SPA-based chromatography matrix." *Id.* (citing Ex. 1002 ¶¶ 88–89). In other words, Petitioner contends that the contacting step is a well-known step in the field of affinity chromatography.

Petitioner contends that "[a]dsorbing target compounds to SPA-based ligands coupled to the solid supports was a well-known and conventional feature of SPA-based affinity chromatography." Pet. 23–24 (citing Ex. 1002 ¶ 122, *see also id.* ¶¶ 120–121; Ex. 1004, 4).

Petitioner contends that "a [person of ordinary skill in the art] would have further understood a 'standard affinity chromatography protocol' would involve the well-known and conventional step of eluting target compounds from SPA-based ligands coupled to the solid supports in a chromatography matrix." Pet. 24 (citing Ex. 1002 ¶ 124, *see also id.* ¶¶ 121–125; Ex. 1004 ¶ 4). Petitioner, therefore contends that it is well-known that a standard affinity chromatography protocol contains three active steps: (1) contacting, (2) adsorbing, and (3) eluting.

Petitioner contends that Linhult teaches a chromatography matrix. Pet. 18. Specifically, Linhult teaches using a HiTrap chromatography affinity column made up of agarose beads that serve as a solid support for "coupling SPA-based ligands." Pet. 18 (citing Ex. 1004, 4; Ex. 1002 ¶ 98). "*Linhult* discloses that its SPA-based ligands were 'coupled to' the solid support agarose beads [] contained in HiTrap™ affinity columns." *Id.* at 18 (citing Ex. 1004, 4). Petitioner contends that "Linhult discloses that 'multimerization' of SPA monomers is performed to 'achieve' an '[SPA-]like' affinity ligand." *Id.* at 19 (citing Ex. 1004, 4; Ex. 1002 ¶¶ 103–106). Petitioner contends that "Figure 1(a), *Linhult* describes at least 55 amino acids of SPA's naturally-occurring C domain (i.e., SEQ ID NO. 1)." *Id.* (citing Ex. 1004, 1, Fig. 1(a); *see* Ex. 1005, Fig. 2; Ex. 1006, Fig. 1; Ex. 1008, 639, Fig. 1). Petitioner contends "that all 'five SPA domains show individual affinity for the Fc-fragment . . . as well as certain Fab-fragments of [antibodies] from most mammalian species." *Id.* at 20 (citing Ex. 1004, 1).

Petitioner contends that it was "known that individual SPA domains, including the C domain, could be used to construct SPA-based affinity ligands for purifying proteins." *Id.* at 20 (citing Ex. 1002. ¶¶ 29, 113; Ex. 1004, 1; Ex. 1006, 12; Ex. 1018 ¶ 29; Ex. 1019, 6:25–34). Petitioner contends that Linhult teaches a person of ordinary skill in the art "that avoiding the Asn_{28}-Gly_{29} dipeptide sequence through a G29A mutation, including on the C domain, would yield an SPA-based ligand having increased alkali-stability." *Id.* at 21 (citing 1002 ¶¶ 112–116; Ex. 1011; Ex. 1012; Ex. 1013). Petitioner acknowledges that "*Linhult* does not expressly disclose a C(G29A)-based SPA ligand," but asserts that "[r]egardless, it would have been obvious to a [person of ordinary skill in the art] to modify *Linhult* based on the teachings of *Abrahmsén* to incorporate a C(G29A)-based SPA ligand in a chromatography matrix." *Id.* at 22 (Ex. 1002 ¶¶ 111–120). Petitioner contends "*Abrahmsén* expressly discloses 'a recombinant DNA coding for ***any of*** the E D A B ***C*** domains of [SPA], wherein the glycine codon(s) in the $Asn_{[28]}$-$Gly_{[29]}$ coding constellation has been replaced by an alanine codon.'" *Id.* at 22 (bracketing in and emphasis in original) (citing Ex. 1005, 2:32–37).

> A [person of ordinary skill in the art] would have had good reason to combine the teachings from *Abrahmsén* with *Linhult* because a G29A mutation was known to increase alkali-stability by avoiding the troublesome Asn_{28}-Gly_{29} dipeptide sequence, i.e., the "most sensitive amino acid sequence to alkaline conditions," such as those used in CIP. (Ex. 1002 ¶¶ 119, 135; Ex. 1004, [2].) Moreover, a [person of ordinary skill in the art] would have been drawn to a C-domain-based ligand, which, as *Linhult* describes, shows individual affinity for antibodies and already includes 23T (as well as 43E), which it disclosed as providing "remarkably increased" stability. (Ex. 1002 ¶¶115-16; Ex. 1004, [1], [8–9].)

Pet. 23.

> Applying the teachings of *Abrahmsén* with *Linhult* would have involved merely combining known elements in the field (e.g., a process for isolating one or more target compounds using an affinity chromatography matrix comprising a G29A-containing ligand coupled to a solid support, as in *Linhult*, and a C(G29A)-based amino acid sequence, as in *Abrahmsén*) according to known ligand-construction methods to yield a predictable result[] (e.g., a process for isolating one or more target compounds using the recited affinity chromatography matrix). (Ex. 1002 ¶136.) *See, e.g., KSR Int'l Co. v. Teleflex, Inc.*, 550 U.S. 398, 415-21 (2007); *Wyers v. Master Lock Co.*, 616 F.3d 1231, 1239-40 (Fed. Cir. 2010).

Pet. 25–26. Petitioner further contends "*Hober*'s disclosure is in the context of SPA-based affinity chromatography utilizing G29A-containing ligands, and, in fact, further confirms that the teachings of *Abrahmsén* are applicable in this context." Pet. 52 (citing Ex. 1002 ¶¶ 260–275; Ex. 1006, 10–12); Ex. 1006, 12 ("the present multimer also comprises one or more of the E, D, A, B, and C domains of Staphylococcal protein A. . . . for structural stability reasons, the glycine residue in position 29 of SEQ ID NOS. 1 has also been mutated, preferably to an alanine residue").

b) Claims 11 and 29

Petitioner contends that "'[t]he capability of "bind[ing] to the Fab part of an antibody,' as recited in claims 11 and 29, is an inherent property of the recited C(G29A)-based SPA ligand." Pet. 30 (citing Ex. 1002 ¶¶ 155–162).

c) Claims 3–10, 12–14, 16–19, 21–28, 30–32, and 34–37

With respect to claims 3–10, 12–14, 16–19, 21–28, 30–32, and 34–37, Petitioner directs our attention to where in the asserted art of record the various limitations of the dependent claims may be found. *See* Pet. 20 n.13, 26–30, 37–41.

2. Patent Owner's Contentions

Patent Owner argues that the Petition fails to demonstrate that there would have been motivation and a reason to make and use the chromatography matrix as claimed (PO Resp. 17–48); that the Petition has not established that there is a reasonable expectation of success in arriving at the claimed matrix (*id.* at 49–54); that the art teaches away from making the G29A modification (*id.* at 38–44); that the artisan would have been motivated make additional mutations (*id.* at 44–48).

a) Matrix

According to Patent Owner, "Petitioners fail to explain *why* the POSA would have been motivated to select Domain C's amino acid sequence as the foundation for an engineered SPA ligand with favorable properties." PO Resp. 19. Specifically arguing that the obviousness analysis requires the prior art be viewed as a whole. PO Resp. 21 (citing *In re Wesslau*, 353 F.2d 238 (CCPA 1965); *In re Enhanced Sec. Rsch., LLC*, 739 F.3d 1347, 1355 (Fed. Cir. 2014); *Impax Lab'ys Inc. v. Lannett Holdings Inc.*, 893 F.3d 1372, 1379 (Fed. Cir. 2018)). In other words, because obviousness requires considering the prior art as a whole and no one was working on domain C at the time of the invention, Patent Owner asserts, it would not have been obvious to select domain C for further development or genetic modification.

Patent Owner argues that because nobody was working on domain C at the time the invention was filed, therefore, the selection of domain C for further development could not possibly be obvious. *See* PO Resp. 24 ("Reliance on *KSR* also is foreclosed by the evidence that no one in the art was seeking to modify Domain C."), *see also id.* at 25 ("But no prior art cited by Petitioners singles Domain C out for further development. Ex. 2025 ¶¶ 89–96"), *id.* at 27 ("Dr. Cramer [Petitioner's expert] himself highlights, it would have been natural for the POSA to further develop the domain—Domain B—that was best understood and for which there was a crystal structure available. Ex. 1002 ¶ 33; Ex. 2015 at 137:20–138:19; Ex. 2017"), *id.* at 28–29 ("The notion that this body of work would lead the POSA to discard the improved ligands the references themselves focus on, and instead start experimenting with mutations to Domain C—strains credulity. Ex. 2025 ¶¶ 92–95").

According to Patent Owner, neither Linhult nor Abrahmsén supply the motivation to start with domain C. "Linhult focuses exclusively on, and concerns improvements to, the alkaline stability of Domain Z by mutating asparagine residues. *See* Ex. 2025 ¶¶ 64–67, 92." PO Resp. 30. "Rather than use Domain C, the POSA reviewing Linhult would be motivated to keep working with Domain Z, adopting the N23T mutation. Ex. 2025 ¶ 67 & n.3." PO Resp. 31. "Neither Abrahmsén itself nor the Petition provide any reason as to why the POSA would have 'plucked' Domain C from among the five listed SPA domains. *WBIP*[*LLC v. Kohler Co.*, 829 F.3d 1317, 1337 (Fed. Cir. 2016)]." PO Resp. 31–32.

b) Reasonable Expectation of Success

Patent Owner argues that "the field of protein engineering is notoriously unpredictable." PO Resp. 22 (citing Ex. 2025 ¶¶ 50–52). Arguing that "despite their supposed structural similarity, there are a number of differences between the naturally-occurring domains of protein A, including five different amino acids in the sequences of Domain B (with which the industry was quite familiar) and Domain C (which remained virtually ignored as of the priority date)." *Id.* at 22–23 (citing Ex. 2025 ¶ 48).

> Protein engineering is a highly complex and unpredictable field and was all the more so as of the priority date more than fifteen years ago. *See*, *e.g.*, Ex. 2025 ¶¶ 50-52. . . . As amply demonstrated by the effect of the G29A mutation on Domain Z's Fab-binding ability, even a single amino acid substitution can drastically alter the properties of a protein. Ex. 2025 ¶ 52; Ex. 2015 at 51:15-52:1 ([Dr. Cramer, Petitioner's expert] agreeing that a single amino acid change can have a significant effect on a ligand's binding ability), 18:10-12, 73:16-20.

PO Resp. 35–36.

Patent Owner argues that

> The Federal Circuit has rejected arguments premised on the notion that a homologous structure renders an invention obvious, particularly given the difficulty and uncertainty in the art as of the priority date. *See, e.g., Amgen, Inc. v. Chugai Pharm. Co.*, 927 F.2d 1200, 1208 (Fed. Cir. 1991) (holding the use of a monkey gene to probe for a roughly 90 percent "homologous"

> human gene would not have been obvious, particularly given expert testimony that isolating a particular gene would have been "difficult" and the lack of certainty in the endeavor).

PO Resp. 37–38. Specifically, Patent Owner argues that the Fab-binding capability of a ligand could not have been predicted and therefore there is no reasonable expectation of success in using the ligand in a process of purifying a target compound. *See* PO Resp. 49–55.

c) Teaching Away

Patent Owner argues that the prior art would have told the person of ordinary skill in the art to avoid a G29A a modification to domain C. PO Resp. 39–44. In other words, it's Patent Owner's contention that the prior art teaches away from making this modification. "The very G29A amino acid substitution Petitioners now suggest the POSA would seek to employ with Domain C would have been known to have rendered Fab binding 'negligible' when implemented in Domain B." PO Resp. 41. Patent Owner argues that a person seeking to improve Fab binding would have avoided a G29A substitution of domain C. PO Resp. 41–44.

d) Additional Modifications

Patent Owner argues that "the prior art would have taught the POSA to make asparagine substitutions, not glycine substitutions, to address alkaline stability concerns." PO Resp. 44–45. In other words, Patent Owner argues the prior art would have suggested making additional substitutions most notably in the asparagine residues of domain C. *Id.* at 46.

3. Petitioner's Reply

In response, Petitioner argues that

> *Abrahmsén* and *Hober* each expressly pointed to a C(G29A) mutation (Ex. 1005, 2:32-37; Ex. 1006, 12), which was known to increase alkali-stability by avoiding the troublesome Asn_{28}-Gly_{29} dipeptide sequence (*see, e.g.*, Ex. 1004, [2]). As the Board recognized, "*Abrahmsén* provides motivation for making [the G29A] mutation in ***any of the IgG binding domains***." (Decision, 26; *see also* Ex. 1057, 97:3-16 (Dr. Bracewell admitting that *Abrahmsén* discloses a G29A mutation to any of the five domains, including Domain C).)

Reply 2.

> A POSA would have reasonably expected success in combining these teachings to achieve the claimed affinity chromatography matrix given the well-known fact that each individual domain, including Domain C, has affinity for antibodies (Ex. 1004, [1]), as well as *Abrahmsén*'s confirmation that G29A "would not interfere with folding [of SPA] or binding to [antibodies]" (Ex. 1057, 99:13-101:21 Ex. 1005, 5:13-16; Ex. 1002 ¶131).

Id. at 3.

Petitioner argues that Patent Owner "has not disputed that *Abrahmsén* disclosed that G29A 'would not interfere with folding to protein A or binding to IgG.' (Ex. 1005, 2:32–37, 5:4–16; Ex. 1057, 109:20–110:17.) Nor does it take issue with its own statements in *Hober* that G29A is advantageous for 'structural stability reasons.' (Ex. 1006, 12.)" *Id.* at 9. Petitioner contends that Patent Owner's lack of binding argument is contradicted by "*Abrahmsén* and *Hober*, which make clear that G29A does not affect the ability of an SPA ligand to bind to an antibody. (Ex. 1005, 2:32–37, 5:4–16; Ex. 1006, 12.)" *Id.* at 10.

Petitioner argues that "a POSA would have started with any one of the naturally occurring domains. (Decision, 26-28.) To then increase alkali stability, a POSA would have made the simplest, well-known substitution: G29A. (Section II.A.1–2; Ex. 1061 ¶¶8–15.)" Reply 11.

Petitioner argues "that Fab-binding was an inherent feature of a C(G29A)-based ligand—which [Patent Owner] does not appear to dispute. (Decision, 34; [Prelim.] Resp., 53–54.) In fact, [Patent Owner] acknowledges that 'C(G29A)-based SPA ligands retained substantial Fab-binding ability.' (Resp., 56–57.)" *Id.* at 14.

Petitioner argues that "Fab-binding is not being used [in the Petition] as part of a finding of a motivation to combine; rather, it is an inherent property [of the composition] being claimed. And necessarily present properties do not add patentable weight when they are claimed as limitations. *In re Kubin*, 561 F.3d 1351, 1357 (Fed. Cir. 2009)." Reply 15–16. Petitioner further argues that Patent Owner's reliance on *Honeywell* is misplaced because "*Honeywell* had to do with an inherent property being used as a teaching in an obviousness analysis; it did not involve a limitation in the challenged claim reciting an inherent property." Reply 15 (citing *Honeywell Int'l Inc. v. Mexichem Amanco Holding S.A. De C.V.*, 865 F.3d 1348, 1355 (Fed. Cir. 2017); *see also Pernix Ireland Pain v. Alvogen Malta Operations*, 323 F. Supp. 3d 566, 607(D. Del. 2018)).

4. Patent Owner's Sur-reply

Patent Owner argues that "Petitioners, and the Institution Decision, overlook an important point of consensus between the parties' experts: the field of protein engineering is notoriously *unpredictable*." Sur-reply 2. Patent Owner maintains that Petitioner has not identified a motivation to start from Domain C. *Id.* at 3. Patent Owner argues that "Petitioners would have the Board look past the multitude of references teaching a preference for Domains B and Z—including Petitioners' foundational references—and seize upon fleeting mentions of Domain C." *Id.* at 7 (citing *In re Wesslau*, 353 F.2d 238, 241 (C.C.P.A. 1965)”). Patent Owner argues that “[m]ere sequence homology does not make the field predictable, as both experts observe, Ex. 2015 at 51:15-52:1, 56:4-12, 75:12-22, 73:16-20; Ex. 2025 ¶¶ 50-52; Ex. 2049 at 72:1-73:12, and as the vastly different Fab-binding properties of the near-identical Domains B and Z well illustrate, Ex. 2029 at 8.” *Id.* at 8–9.

Patent Owner argues that “Abrahmsén’s computer simulation was of unmodified Protein A as a whole, not a Domain C (or G29A-modified) monomer or multimer, and thus does not reveal the impact of a G29A mutation on protein folding or IgG affinity. Ex. 2025 ¶ 103; Ex. 2049 at 131:7–10.” Sur-reply 11.

5. Analysis

a) Claims 1, 2, and 20

Independent claims 1, 2, and 20 of the ’007 patent are directed to a method of isolating a target compound using a chromatography matrix composition. Ex. 1001, 15:40–64, 17:22–40. The claims recite three active steps: (1) contacting, (2) adsorbing, (3) eluting the target compound from the chromatography matrix, and (4) cleaning the chromatography matrix in place using a NaOH solution. *Id.* Claim 1 further stipulates that the chromatography matrix composition (a solid support) has the following features: a ligand is attached to the matrix and the ligand is made up of at least two polypeptides comprising 52 contiguous amino acids of SEQ ID NO: 1[6] each having a G29A mutation. Claim 2 depends from claim 1 and

[6] Wild type amino acid sequence of domain C from *Staphylococcus* protein A (SPA). *See* Ex. 1001, 4:27, 6:35–36, 6:51–52.

recites "wherein the amino acid sequence of each polypeptide comprises at least 55 contiguous amino acids of a modified SEQ ID NO. 1." *Id.* at 15:62–64; Pet. 20 n.13. Claim 20 is similar to claim 1, except that claim 20 recites "at least 55 amino acids in alignment with SEQ ID NO. 1" instead of "at least 52 contiguous amino acids of modified SEQ ID NO. 1" as recited in claim 1, and does not recite the eluting step. *Id.* at 17:22–40.

(1) *Method*

Claims 1, 2, and 20 are directed to a method of using a particular matrix. The method comprises two parts: (a) the process steps and (b) the structure of the matrix.

(a) Process Steps

Petitioner asserts that the combination of Linhult, Abrahmsén, and Hober teaches or suggests the standard affinity chromatography process steps of (1) contacting, (2) adsorbing, (3) eluting, and (4) cleaning in place for the reasons set forth in the Petition. Pet. 16–31; Ex. 1002 ¶ 24.

Linhult teaches making affinity chromatography columns with protein Z, Z(F30A), and other mutated variants. These modified proteins were covalently attached to HiTrap columns in Linhult using NHS-chemistry. Ex. 1004, 4. Linhult uses an affinity matrix column to isolate IgG and measures the loading capacity of the column after repeated cleaning in place (CIP) cycles. Ex. 1004, 4. In Linhult's studies, human polyclonal IgG was prepared and injected onto the columns in excess and "[a] standard affinity chromatography protocol was followed." *Id.* at 4. We find that Linhult's loading of IgG onto the column satisfies the contacting step as recited in the claims. Ex. 1004, 4, *see also id.* ("The columns were pulsed with TST

(25mMTris-HCl pH 7.5, 150 mM NaCl, 1.25 mM EDTA, 0.05% Tween 20) and 0.2 M HAc, pH 3.1. Human polyclonal IgG in TST was prepared and injected onto the columns in excess. A standard affinity chromatography protocol was followed for 16 cycles.”). Linhult teaches that “the amount of eluted IgG was measured after each cycle to determine the total capacity of the column.” *Id.* Linhult thereby expressly teaches the contacting and eluting steps, and following standard chromatography protocols the adsorbing step is implied. *Id.*; Ex. 1002 ¶ 24 (“After loading is completed, an additional step to wash out certain remaining impurities is employed. ([Ex. 1006 at 15–17]). Following the loading and wash steps, a different solution, typically one of low pH, is applied onto the column to elute the antibody”). Linhult also teaches a cleaning in place step using an alkaline cleaning agent. “The cleaning agent was 0.5 M NaOH and the contact time for each pulse was 30 min, resulting in a total exposure time of 7.5 h for Z(F30A) and mutants thereof.” Ex. 1004, 4.

Abrahmsén teaches using an IgG bound column for purifying the dimeric Z fragment from a supernatant. Ex. 1005, 9:60–10:15; Ex. 1002 ¶¶ 64–68. Abrahmsén affinity purification protocol is as follows:

> The supernatant was passed through the column at a speed of 12 ml/h and the amount of IgG binding material [i.e. the Z fragment] was analyzed before and after it was run through the column. The bound material was washed with TS [(150 mM NaCl 50 mM tris HC pH 7.5)] supplemented with 0.05% Triton X-100 and then TS and finally with 0.05 M ammonium acetate before elution with 1M acetic acid pH adjusted to 2.8 with ammonium acetate.

Ex. 1005, 10:8–16. The contacting step in Abrahmsén’s protocol occurs when the “supernatant was passed through the column,” the adsorbing step occurs before or during the time “[t]he bound material was washed,” and "elution" step occurs when the column is treated "with 1M acetic acid pH adjusted to 2.8 with ammonium acetate." *Id.* Although Abrahmsén column has IgG bound to the column, instead of an SPA domain, the reference still teaches the common chromatography steps of (1) contacting, (2) adsorbing, and (3) eluting a target molecule.

Hober teaches that "protein monomers can be combined-into multimeric proteins, such as dimers, trimers, tetramers, pentamers etc." Ex. 1006, 11. These monomer units can be linked with stretches of amino acids ranging from 0–15 amino acids. *Id.* Hober teaches

> a matrix for affinity separation, which matrix comprises ligands that comprise immunoglobulin-binding protein coupled to a solid support, in which protein at least one asparagine residue has been mutated to an amino acid other than glutamine. . . The mutated protein ligand is preferably an Fc fragment-binding protein, and can be used for selective binding of IgG. . . the ligands present on the solid support comprising a multimer.

Ex. 1006, 13. Hober describes a typical chromatographic run cycle consisting of: sample application of 10 mg polyclonal human IgG; extensive washing-out of unbound proteins; elution at 1.0 ml/min with elution buffer; followed by Cleaning-In-Place (CIP) with CIP-buffer with a contact time between column matrix and 0.5 M NaOH of 1 hour. *Id.* at 37.

Patent Owner does not dispute that the references disclose the recited chromatography process of contacting, adsorbing, eluting, and cleaning in place. *See generally* PO Resp.; *see id.* at 7 (citing Ex. 1002 ¶ 24; Ex. 2025 ¶ 43).

(b) Chromatography Matrix

The dispute between the parties is whether a person of ordinary skill in the art would have modified the process disclosed in Linhult using the

G29A modified SPA C domain as disclosed in Abrahmsén. Pet. 22 ("Abrahmsén unequivocally discloses performing a G29A mutation on SPA's C domain."); *See* PO Resp. 17–54.

Linhult discloses a process for isolating one or more target compound(s) using chromatography matrices (solid support) comprising SPA ligands. Pet. 16–32. Linhult explains that SPA is a cell surface protein expressed by *Staphylococcus aureus* and consists of five highly homologous domains (E, D, A, B, and C). Ex. 1004, 1. Each of "[t]he five SPA domains show individual affinity for the Fc-fragment [11 residues of helices 1 and 2 (domain B)], as well as certain Fab-fragments of immunoglobulin G (IgG) from most mammalian species." *Id.* (bracketing in original). "Due to the high affinity and selectivity of SPA, it has a widespread use as an affinity ligand for capture and purification of antibodies." Ex. 1004, Abstr., *see also id.* at 1 ("SPA has a widespread use in the field of biotechnology for affinity chromatography purification, as well as detection of antibodies.").

Linhult explains that, in column chromatography, sodium hydroxide (NaOH) is probably the most extensively used cleaning agent for removing contaminants such as nucleic acids, lipids, proteins, and microbes, and a CIP step is often integrated in the protein purification protocols using chromatography columns. Ex. 1004, 1. "Unfortunately, protein-based affinity media show high fragility in this extremely harsh environment, making them less attractive in industrial-scale protein purification. SPA, however, is considered relatively stable in alkaline conditions." *Id.* at 2. Linhult explains that the combination of asparagine with a succeeding glycine is the most sensitive amino acid sequence to alkaline conditions. *Id.* Linhult teaches that "[a]n exchange of glycine 29 for an alanine has been made in order to avoid the amino acid combination asparagine–glycine, which is [sensitive to alkaline conditions and is also] a cleavage site for hydroxylamine." *Id.*

Petitioner's declarant, Dr. Cramer avers that the "Z" domain referenced in Linhult refers to a synthetic version of the wild-type (i.e., natural) B domain of SPA, in which the naturally occurring glycine in the Asn_{28}-Gly_{29} dipeptide sequence is replaced by an alanine residue to create an Asn_{28}-Ala_{29} dipeptide sequence. Ex. 1002 ¶ 30 (citing Ex. 1004, 2, Fig. 1(a); Ex. 1007, 3, Fig. 1); ¶ 31 (citing Ex. 1005). We credit Petitioner's declarant, Dr. Cramer for establishing that the C domain sequence disclosed in Linhult contains 55 amino acids in SEQ ID NO: 1 as claimed. Ex. 1002 ¶ 109 (showing a sequence alignment), *see also id.* ¶ 322 (showing sequence alignment of domain C of Abrahmsén with SEQ ID NO: 1).

According to Abrahmsén, the IgG binding domains E, D, A, B, and C of SPA were known. *See* Ex. 1005, 3:25–35, 4:34–37, Fig. 2. Relying on "computer analysis [Abrahmsén] surprisingly showed that the Gly in the Asn-Gly dipeptide sequence could be changed to an Ala. This change was not obvious as glycines are among the most conserved amino acids between homologous protein sequences due to their special features." *Id.* at 5:7–9. Abrahmsén teaches that in a preferred embodiment, "the glycine codon in the Asn-Gly constellation has been replaced by an alanine codon." *Id.* at 2:21–23. Thus, Abrahmsén provides motivation for making this mutation in any of the IgG binding domains E, D, A, B, and C of Staphylococcal protein A. Abrahmsén teaches recombinant DNA fragments coding for any of the E, D, A, B, and C domains of Staphylococcal protein A, wherein the glycine codon(s) in the Asn-Gly coding constellation have been replaced by an alanine codon. *Id.* at 2:33–37.

Abrahmsén, like Linhult, exemplifies the cloning of and expression of the Z-fragment. Ex. 1005, 7:65–10:56. Abrahmsén teaches that "the Z-region is the part of the Z-fragment coding for the IgG binding domain." *Id.* at 3:39–41. Abrahmsén purifies the recombinant Z protein using an IgG column. *Id.* at 10:26–28. In one embodiment, Abrahmsén provides "a recombinant DNA sequence comprising at least two Z-fragments" in which "the number of such amalgamated Z-fragments is preferably within the range 2–15, and particularly within the range 2–10." *Id.* at 2:27–31. Abrahmsén, therefore, reasonably suggests making multimeric constructs. Abrahmsén also uses column chromatography to purify a Z domain containing protein.

Hober teaches a multimer ligand that

> also comprises one or more of the E, D, A, B, and C domains of *Staphylococcal* protein A. In this embodiment, it is preferred that asparagine residues located in loop regions have been mutated to more hydrolysis-stable amino acids. In an embodiment advantageous for structural stability reasons, the glycine residue in position 29 of SEQ ID NOS. 1 has also been mutated, preferably to, an alanine residue. Also, it is advantageous for the structural stability to avoid mutation of the asparagine residue in position 52, since it has been found to contribute to the a-helical secondary structure content of the protein A molecule.

Ex. 1006, 12, *see also id.* at 9 ("SEQ ID NO 1 defines the amino acid sequence of the B-domain of SpA").

Here, the teachings of Linhult, Abrahmsén, and Hober suggest mutating the glycine at position 29 for an alanine in any one of the IgG binding domains of E D A B or C of SPA in order to avoid protein degradation. Ex. 1004, 2; Ex. 1005, 5:4–9. We, therefore, agree with Petitioner that the art expressly suggests that the glycine codon can be mutated for an alanine codon in any one of the SPA IgG binding domains E, D, A, B, or C. Pet. 23 (citing Ex. 1002 ¶¶ 99–111). Attaching any one of SPA mutated IgG binding domains E D A B or C to a matrix using "known ligand-construction methods to yield a predictable result[] (e.g., the claimed affinity chromatography matrix)" would have been obvious. Pet. 25 (citing Ex. 1002 ¶ 130). As the Federal Circuit has explained, "[w]here a skilled artisan merely pursues 'known options' from 'a finite number of identified, predictable solutions,' the resulting invention is obvious under Section 103." *In re Cyclobenzaprine Hydrochloride Extended-Release Capsule Patent Litig.*, 676 F.3d 1063, 1070 (Fed. Cir. 2012) (quoting *KSR*, 550 U.S. at 421).

Accordingly, we agree with Petitioner that the combination of Linhult Abrahmsén, and Hober expressly suggests mutating the glycine codon for an alanine codon in *any one* of the SPA IgG binding domains E, D, A, B, or C. Pet. 23 (citing Ex. 1002 ¶¶ 115–116). We also agree that the cited art teaches using these mutant SPA domains in column chromatography for the isolation of antibodies.

We address Patent Owner's contentions below.

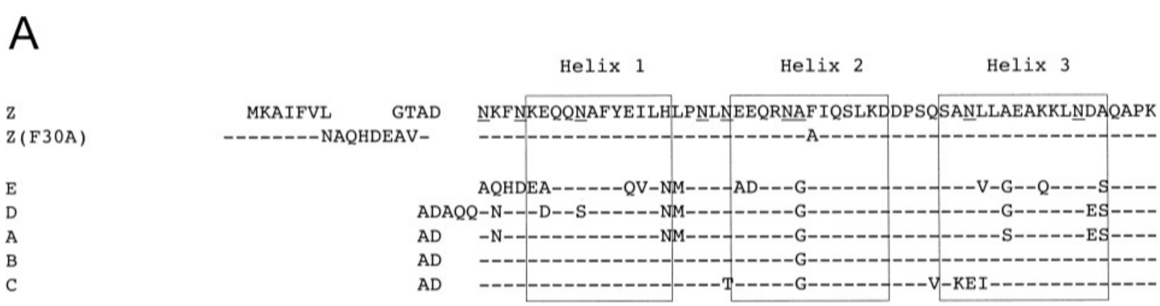
(2) *Response*

(a) Matrix

We do not find Patent Owner's argument that the Petition fails to identify a reason to select domain C persuasive. PO Resp. 17–38. Specifically, we are not persuaded by Patent Owner's contention that just because nobody was working on domain C at the time the invention was filed means selection of domain C would not have been obvious. *See* PO

Resp. 32 ("A general recognition that there exist five naturally occurring protein A domains is not a motivation to use each of them as a starting point for the claimed mutations").

Petitioner's articulated obviousness ground is premised on the knowledge that *any one* of the five SPA IgG binding domains are known to bind IgG and can function as a ligand for the purification of antibodies. Linhult and Abrahmsén both expressly suggest that the glycine codon at position 29 can be mutated for an alanine codon in *any one* of the SPA IgG binding domains E, D, A, B, or C. Ex. 1004, 2; Ex. 1005, 2:32–37. Here, the SPA IgG binding domains comprise a short list of 5 members: E, D, A, B, or C. Of these 5 members, the glycine at position 29 in domain B has already been mutated to an alanine to create a domain Z which has been shown to retain IgG binding activity. Ex. 1004, 6 (Fig. 3). Figure 1A of Linhult is reproduced below.



Linhult's Figure 1A, reproduced above, shows the amino acid alignments of the Z, Z(F30A) and the five homologous domains (E, D, A, B, and C). The three boxes show the α-helices. Ex. 1004, 2; Ex. 1005, Fig. 2.

Here, Linhult and Abrahmsén show that the IgG binding domains of SPA – E, D, A, B, or C share many structural similarities. *See* Ex. 1004, 2 (Fig. 1(a) (reproduced above)); Ex. 1005, 3:25–35. As discussed in our Institution Decision (Dec. 30–31), there are a finite number – five (5) – SPA

IgG binding domains and each possesses the dipeptide sequence Asp-Gly known to a be a target for alkaline protein degradation. Therefore, the solution of mutating the glycine at position 29 for an alanine to remove the alkaline sensitive sequence is not a product of innovation but of ordinary skill and common sense. *See Wm. Wrigley Jr. Co. v. Cadbury Adams USA LLC*, 683 F.3d 1356, 1364-65 (Fed. Cir. 2012) (quoting *KSR*, 550 U.S. at 421). It is well established that

> [s]tructural relationships may provide the requisite motivation or suggestion to modify known compounds to obtain new compounds. For example, a prior art compound may suggest its homologs because homologs often have similar properties and therefore chemists of ordinary skill would ordinarily contemplate making them to try to obtain compounds with improved properties.

In re Deuel, 51 F.3d 1552, 1558 (Fed. Cir. 1995).

There is also an express teaching in both Linhult and Abrahmsén to mutate the glycine at position 29 to an alanine in order to prevent degradation of the protein and increase stability, which further supports the obviousness of incorporating the mutation into any one of the IgG binding domains that has the Asn-Gly dipeptide. *See, e.g.*, *SIBIA Neurosciences, Inc. v. Cadus Pharm. Corp.*, 225 F.3d 1349, 1358–59 (Fed. Cir. 2000) (stating that an express teaching in the prior art suggesting a particular modification establishes obviousness).

Because the G29A modification would have provided ligands that are less susceptible to alkaline conditions and are resistant to hydroxylamine cleavage, Petitioner has provided a sufficient evidence-backed reason for making the modification in any one of the domains. Pet. 16–25; Reply 3–5; Ex. 1061 ¶¶ 8–11; Ex. 1004, 2; Ex. 1005, 2:32–37.

(b) Reasonable Expectation of Success

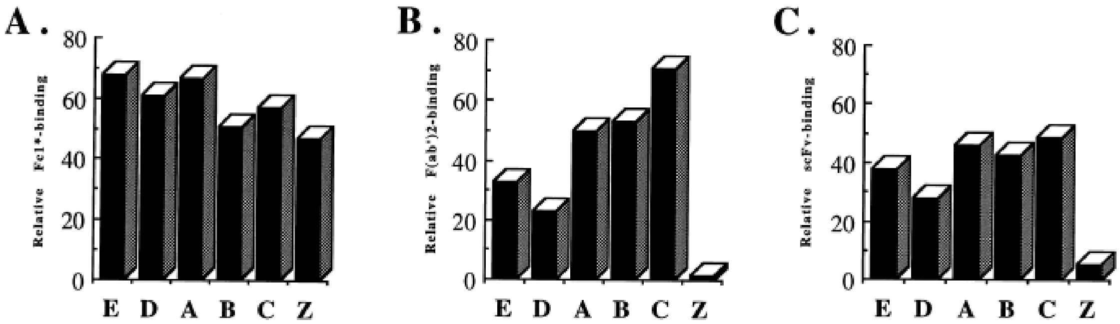
We are not persuaded by Patent Owner's contention that there is no reasonable expectation of success in using a G29A mutation in domain C. PO Resp. 49–55.

Linhult explains that removing the Asp–Gly amino acid combination not only results in the removal of the hydroxylamine cleavage site but also creates a product that is more alkaline resistant. *See* Ex. 1004, 2 ("An exchange of glycine 29 for an alanine has been made in order to avoid the amino acid combination asparagine–glycine, which is a cleavage site for hydroxylamine. Asparagine with a succeeding glycine has also been found to be the most sensitive amino acid sequence to alkaline conditions.").

Abrahmsén teaches that this Asn-Gly amino acid combination is present in all five SPA IgG binding domains and that mutating the dipeptide would not interfere with IgG binding. Ex. 1005, 4:56–58 ("The Asn-Gly dipeptide sequence is sensitive to hydroxylamine. As this sequence is kept intact in all five IgG binding domains of protein A. . . . However, by simulating the Gly to Ala amino acid change in the computer we concluded that this change would not interfere with folding to protein A or binding to IgG."). Abrahmsén's conclusion that the mutation would not interfere with biding to IgG is supported by Abrahmsén (*see* Ex. 1005, 9:60–10:35), Linhult (*see* Ex. 1004, 6 (Fig. 3)), and Jansson (Ex. 2029).[7] Jansson's Fig. 3,

[7] Patent Owner cites Jansson (Ex. 2029) for the position that domain Z has negligible binding to Fab. *See* PO Resp. 40–41. However, claims 1, 2, and 20 are not limited to Fab binding. Indeed the claims do not even require IgG binding.

reproduced below.



Jansson Figure 3 (Panel A), reproduced above, shows a side-by-side comparison of Fc1*, Fab, and scFv binding to SPA domains. The figure shows that the single G29A mutation between domain B and domain Z results in a protein that is able to bind IgG.[8] Comparing panel A - domain B domain with panel A - domain Z, the relative binding remains close to 50%, indicating that the G29A mutation between domains B and C does not interfere with IgG binding. Ex. 2029, 6. This is a result predicted by Abrahmsén's computer modeling and substantiated by Abrahmsén domain Z purification and Linhult's IgG purification. *See* Ex. 1005, 4:56–58, 9:60–10:35; Ex. 1004, 6 (Fig. 3).

"Obviousness does not require absolute predictability of success . . . all that is required is a reasonable expectation of success." *In re Droge*, 695 F.3d 1334, 1338 (Fed. Cir. 2012) (quoting *In re Kubin*, 561 F.3d 1351, 1360

[8] Fc1* is the constant region of human IgG1. Ex. 2029, 4. Fc1* is understood to be used as the "IgG control" in Jansson. Patent Owner's counsel explains that "Part A is Fc binding. So that is, I believe the way they did this experiment was with Fc fragments, but it's generally acknowledged, you know, these antibodies all have an Fc domain if they're a whole antibody and that reflects the fact that all of these domains A, B, C, D and E and domain Z, which is B with the G29A mutation, retain this Fc binding." Tr. 70:6–11.

(Fed. Cir. 2009) (citing *In re O'Farrell*, 853 F.2d 894, 903–04 (Fed.Cir.1988)); *Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359 (Fed. Cir. 2016) (explaining that the expectation of success issue involves a showing of "a reasonable expectation of achieving *what is claimed*") (emphasis added). "Scientific confirmation of what was already believed to be true may be a valuable contribution, but it does not give rise to a patentable invention." *Pharma Stem Therapeutics, Inc. v. ViaCell, Inc.,* 491 F.3d 1342, 1363–1364 (Fed. Cir. 2007).

Here, the record supports that each individual SPA domain, including the C domain, has affinity for IgG antibodies. Ex. 1004, 1 ("The five SPA domains show individual affinity for the Fc-fragment [11 residues of helices 1 and 2 (domain B)], as well as certain Fab-fragments of immunoglobulin G (IgG) from most mammalian species." (bracketing in original)). Abrahmsén suggests making a mutation of Asn-Gly coding constellation in *any one* of the SPA domains by replacing a glycine codon with an alanine codon to remove the Asn-Gly dipeptide sequence known to be sensitive to hydroxylamine degradation. *See* Ex. 1005, 4:56–5:16, *see also id.* Fig. 2 (showing the Asn-Gly coding constellation in all SPA domains); Ex. 1006, 2 ("the shortest deamidation half times have been associated with the sequences –asparagine–glycine and – asparagine–serine"). Abrahmsén's confirms that a G29A mutation on SPA would not interfere with folding of SPA protein and the binding to antibodies. Ex. 1005, 5:13–16 ("by simulating the Gly to Ala amino acid change in the computer we concluded that this change would not interfere with folding to protein A or binding to IgG."), 9:60–10:35 (using IgG columns to purify protein Z dimers). Abrahmsén's computer modeling suggests that IgG binding is not impacted by the mutation and this is confirmed by Linhult's experiments showing that the G29A mutant of domain B (a.k.a. domain Z) binds IgG. Ex. 1004, 6 (Fig. 3); *see also* Ex. 1005, 9:60–10:35.

Patent Owner argues that "Abrahmsén's computer simulation was of unmodified Protein A as a whole, not a Domain C (or G29A-modified) monomer or multimer, and thus does not reveal the impact of a G29A mutation on protein folding or IgG affinity. Ex. 2025 ¶ 103; Ex. 2049 at 131:7-10." Sur-reply 11.

We are not persuaded by Patent Owner's contention that the information gained by computer modeling of the SPA native domain B – IgG crystal structure could not be extrapolated to other SPA domains that are structurally very similar.

As Petitioner's expert, Dr. Cramer explains

> [i]t was well known that the researchers who developed the Z domain based on the wild-type B domain (rather than any of the other four SPA domains) did so for two reasons. (*See*, *e.g.*, Ex. 1007 at 109.) First, a crystal structure of the wild-type B domain binding to an antibody happened to be available in 1981 for analysis. (*See*, *e.g.*, *id.*; Ex. 1005 at col. 4:56-68; Ex. 1017.) And, second*, their work would be informative of mutations that could be done on all five of the highly homologous SPA domains more generally*. (*See*, *e.g.*, Ex. 1005 at col. 2:32-37; Ex. 1007 at 109; Ex. 1008 at 639, Fig. 1.)

Ex. 1002 ¶ 33 (emphasis added). Dr. Cramer further explains that "[t]hey did the computer modeling based on that complex because that's the crystal structure that they had. It wasn't done because the B domain is special. . . . And then there's several other places where they state clearly that they could also do the other domains with expected similar results." Ex. 2015, 138:8–22.

We agree with Petitioner that the teachings of Linhult, Abrahmsén, and Hober provide a reasonable expectation of success at arriving at a chromatography composition that contains the SPA domain C ligand with the G29A mutation that can be used in a process for purifying IgG. Pet. 24 (citing Ex. 1004, 4; Ex. 1002 ¶¶ 124–126), *see id.* at 48–49 (citing Ex. 1006, 10–12; Ex. 1002 236–263).

(c) No Teaching Away

We are also not persuaded by Patent Owner's contention that the art teaches away from the G29A substitution because it interferes with Fab binding. *See* PO Resp. 39–42; Sur-reply 9; Ex. 2009 at 2; Ex. 2010 at 25; Ex. 2012 at 25–26; Ex. 2029 at 7.

None of claims 1, 2, or 20 recite a need to bind the Fab region of an antibody or that the target molecule is Fab. All that is required by these claims is that they adsorb a target molecule and that you can elute the target molecule from the matrix. The target molecule, therefore, can reasonably encompass IgG.

Petitioner's articulated rationale is that there was an expectation that the composition binds antibodies, including monoclonal antibodies, and therefore, would be useful in a process of isolating antibodies. Petitioner contends that:

> A POSA would have also reasonably expected such a combination to achieve a process for isolating one or more target compounds using the recited affinity chromatography matrix given the well-known fact that each individual SPA domain, including the C domain, has affinity for antibodies (Ex. 1004, [1]) as well as *Abrahmsén's* confirmation that a G29A mutation on SPA "would not interfere with folding [of SPA] or binding to [antibodies]" (Ex. 1005, 5:13-16). (Ex. 1002 ¶137.)

Pet. 26; *see also* Reply 8 ("'A POSA would not have been motivated only by Fab-binding ability, as even Dr. Bracewell agreed that 'a POSA would have understood that it was desirable to purify ***monoclonal antibodies*** for therapeutic use in 2006.' (Ex. 1057, 75:17-76:4, 113:23-114:11, 157:24-158:9; Ex. 1061 ¶29)").

The law does not require that the teachings of the reference be combined for the reason or advantage contemplated by the inventor, as long as some suggestion to combine the elements is provided by the prior art as a whole. *In re Beattie*, 974 F.2d 1309, 1312 (Fed. Cir. 1992); *In re Kronig*, 539 F.2d 1300, 1304 (CCPA 1976); s*ee In re Kemps*, 97 F.3d 1427, 1430 (Fed. Cir. 1996) ("[T]he motivation in the prior art to combine the references does not have to be identical to that of the applicant to establish obviousness.").

Here, Linhult teaches that "[t]he five SPA domains show individual affinity for the Fc-fragment [11 residues of helices 1 and 2 (domain B)], as well as certain Fab-fragments of immunoglobulin G (IgG) from most mammalian species." Ex. 1004, 1 (bracketing in original) (citation omitted). Linhult, therefore, teaches that *any one* of the SPA IgG binding domains E, D, A, B, or C can bind the Fc region of an antibody and can therefore be used as a ligand for purifying IgG antibodies. In addition, the combination of Linhult and Abrahmsén suggests making the G29A mutation in each of the domains because it would provide ligands that are less susceptible to protein degradation. Ex. 1004, 2; *see also* Ex. 1005, 2:33-37 ("[A] recombinant DNA fragment coding for any of the E D A B C domains of staphylococcal protein A, wherein the glycine codon(s) in the Asn-Gly coding constellation has been replaced by an alanine codon.").

Patent Owner contends that the G29A mutation would lead to a reduction in the Fab binding of domain C, and therefore, would lead away from making the mutation. PO Resp. 40–41 (citing Ex. 2010, 2; Ex. 2011, 25; Ex. 2012, 25–26; Ex. 2013, 2–3; Ex. 2025 ¶¶ 105–109; Ex. 2029, 6–7). Patent Owner's cited references are directed to Fab binding. But claims 1, 2, and 20 are not limited to Fab binding. Showing that the G29A mutation interferes with Fab binding does not teach away from mutating any of SPA domains E, D, A, B, or C in order to create an IgG binding ligand that retains its' binding affinity for Fc yet is less sensitive to cleaning in place solutions. *See, e.g.*, Ex. 2013, 3 ("The site responsible for Fab binding is structurally separate from the domain surface that mediates Fcɣ[[9]] binding."). Accordingly, we are not persuaded by Patent Owner's contention that with respect to claims 1, 2, and 20 that the art teaches away from Petitioner's proposed combination.

(d) Additional Modifications

We also disagree with Patent Owner's contention that the ordinary artisan would not stop with a single G29A mutation in a SPA domain. *See* PO Resp. 44–48. Here, Abrahmsén expressly suggests making only a single mutation. Specifically, Abrahmsén contemplates "a recombinant DNA fragment coding for any of the E D A B C domains of staphylococcal protein A, wherein the glycine codon(s) in the Asn-Gly coding constellation

[9] Fcɣ is the constant region of IgG involved in effector function. Specifically, "[t]he Fcɣ binding site has been localized to the elbow region at the CH2 and CH3 interface of most IgG subclasses, and this binding property has been extensively used for the labeling and purification of antibodies." Ex. 2013, 1. In other words, Fcɣ and Fc terminology are used interchangeably in the art.

has been replaced by an alanine codon" without additional mutations. Ex. 1005, 2:33–37.

(3) *Summary*

We find that Petitioner has shown by a preponderance of the evidence that the combined teachings of at least Linhult and Abrahmsén suggests the use of *any one* of the SPA IgG binding domains E, D, A, B, or C as the starting ligand for purifying IgG antibodies, and that making the G29A mutation in *any one* of the domains would have been obvious because it would have provided ligands that are less susceptible to alkaline conditions and are resistant to hydroxylamine cleavage. Pet. 16–25; Reply 3–5; Ex. 1061 ¶¶ 8–11; Ex. 1004, 2; Ex. 1005, 2:32–37.

Having considered the evidence and argument cited in the Petition, which we have described above and find persuasive, we are persuaded that Petitioner has shown by a preponderance of evidence of record that the combination of Linhult, Abrahmsén, and Hober teaches each of the limitations of claims 1, 2, and 20. Petitioner not only has articulated a sufficient motivation for making the combination but has also established that there is a reasonable expectation of success for the binding of an IgG antibody to a SPA domain that contains an G29A mutation.

b) Claims 11 and 29

Petitioner argues that "[t]he 'capab[ility] of binding to the Fab part of an antibody,' as recited in claims 11 and 29, is an inherent property of the recited C(G29A)-based SPA ligand." Pet. 30 (citing Ex. 1002 ¶¶ 155–162). Petitioner contends that a person of ordinary skill in the art did not need to recognize the Fab binding property of Domain C to be motivated to select that domain for modification. "A POSA would not have been motivated only by Fab-binding ability,[] as even Dr. Bracewell [Patent Owner's expert] agreed that 'a POSA would have understood that it was desirable to purify ***monoclonal antibodies*** for therapeutic use in 2006.'" Reply 8 (citing Ex. 1057, 75:17–76:4, 113:23–114:11, 157:24–158:9; Ex. 1061 ¶ 29).

Patent Owner argues that

> [t]he very G29A amino acid substitution Petitioners now suggest the POSA would seek to employ with Domain C would have been known to have rendered Fab binding "negligible" when implemented in Domain B. Ex. 2009 at 2; *see also, e.g.*, Ex. 2010 at 2 ("Fab binding activity is located to a region determined by helices 2-3, including the position mutated to yield the Z domain."); Ex. 2011 at 25 ("[I]t only takes a single residue change in SpA to eliminate either Fab or Fc binding. The sole difference in domain Z compared to domain B is the substitution of a glycine to an alanine . . . ."); Ex. 2012 at 25-26 ("[D]omain Z containing a single G29A-substitution compared to domain B exhibits little or no [Fab] binding. This might be due to the substitution since the C_β of the alanine would perturb the interaction between the two molecules.").

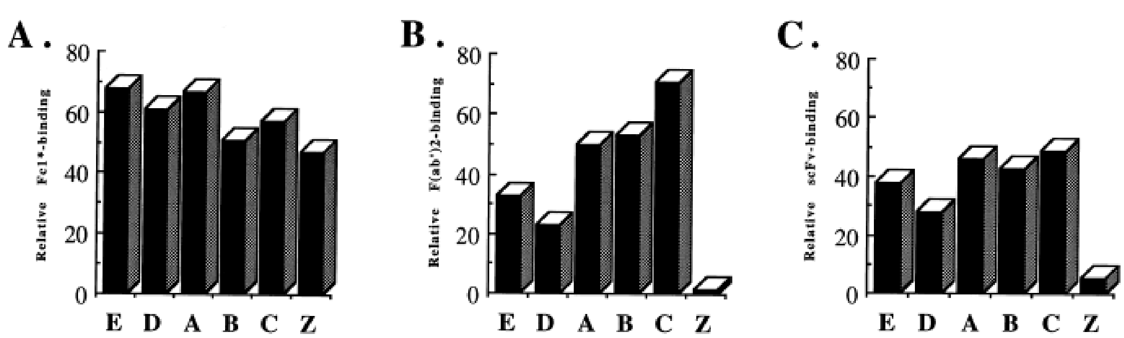
PO Resp. 41.

Because claim 11[10] is directed to a "[a] process for isolating one or more target compound(s)" and identifies that the target compound is "the Fab part of an antibody" (Ex. 1001, 16:66–67) the evidence needs to show a reasonable expectation that a mutated SPA ligand binds Fab. Without such a showing, there is no reasonable expectation that the process would result in the purification of a Fab target. In other words, because the claims are process claims the Petitioner needs to establish that a mutated SPA domain would reasonably bind a Fab fragment.

[10] Claim 11 and 29 recite similar limitations.

We agree with Petitioner, that Linhult establishes that a G29A mutation in domain B (resulting in domain Z) does not interfere with IgG binding. *See* Ex. 1004, 6 (Fig. 3 (showing IgG binding with domain Z). Linhult, however, is silent with respect to Domain Z's ability to bind to Fab fragments. *See generally* Ex. 1004. Abrahmsén similarly establishes Domain Z binding to IgG but is also silent with respect to Domain Z binding Fab. *See generally* Ex. 1005. Hober also does not disclose Fab binding of a mutant SPA domain. *See generally* Ex. 1006. Thus, each of Linhult, Abrahmsén, or Hober are silent with respect to Fab binding to a mutated SPA domain.

Jansson (Ex. 2029), cited by Patent Owner, supports the position that Fab binding to mutated SPA domains is unpredictable. Jansson, just like Linhult, recognizes that "[a]ll [SPA] domains bound to a recombinant human IgG1 Fc fragment with similar strength. For the first time, binding to human Fab was demonstrated for all *native SPA domains*, using both polyclonal $F(ab')_2$ and a recombinant scFv fragment as reagents." Ex. 2029, Abstract (emphasis added). Jansson, however, establishes that "the engineered Z domain showed a considerably lower affinity for Fab as compared to the native domains." *Id.* Jansson Fig. 3, reproduced below, shows that the G29A mutation results in a loss of Fab binding ability.



Jansson Figure 3, reproduced above, shows the side-by-side comparison of Fc1*[11], Fab, and scFv binding and confirms what was already suggested in Linhult, Abrahmsén, and Hober – that a composition containing the G29A mutation in a SPA domain can bind IgG. Ex. 2029, 6 (Fig. 3 (*compare* Panel A-domain B, *with* Panel A-domain Z). Panel B in Jansson Figure 3, however, shows that the single G29A mutation between Domain B and Domain Z results in the loss of Fab binding. Ex. 2029, 6 (Fig. 3 (*compare* Panel B-Domain B, *with* Panel B-Domain Z). At the time the invention was made it was also known that "[t]he site responsible for Fab binding is structurally separate from the domain surface that mediates Fcɣ binding." Ex. 2013, 3. Thus, on this record, establishing that a mutation that does not interfere with IgG binding says nothing about the ability of a mutated SPA domain to bind Fab.

We, therefore, agree with Patent Owner's contention that based on the prior art, the Fab binding capacity was unknown with the modification as suggest by the combination of Linhult, Abrahmsén, and Hober.[12] Patent Owner has provided evidence that Fab biding capacity of a mutated SPA

[11] Fc1* is the constant region of human IgG1. Ex. 2029, 4. Fc1* is understood to be used as the IgG control in Jansson. Fc1* is functionally equivalent to Fcɣ in SPA binding.

[12] In *JSR Corporation et al. v. Cytiva Bioprocess R&D AB et al.*, IPR2022-00036, Paper 41 at 44–49 (PTAB April 19) (Final Written Decision) we determined that the capability of SEQ ID NO:1 to bind Fab is an inherent feature of the structure claimed. The present claims, however, are directed to a method of isolating Fab which requires prior knowledge that the ligand binds Fab. Fab is a digestion product of a whole IgG molecule treated with papain and is not naturally found in IgG samples. *See above* I.F. In other words, when isolating IgG with a column containing SEQ ID NO: 1 there would be no elution of Fab because the fragments are not present in an IgG containing sample.

domain protein is unpredictable. *See* PO Resp. 56–57 (Ex. 2029, 5–6). The disclosures in the prior art, therefore, support Patent Owner's position that "the SPA ligands of the claimed chromatography matrices unexpectedly retained their ability to bind to the Fab part of an antibody despite the substitution of an alanine for the glycine at position 29 of the Domain C sequence." PO Resp. 57 (citing Ex. 2025 ¶ 123; Ex. 2030, 18–19).

Because the art does not support the conclusion that G29A mutation in a SPA domain ligand binds Fab, Petitioner has not established by a preponderance of the evidence of record that the process of isolating Fab target using a mutated SPA domain C ligand as required by claims 11 and 29 would have been obvious based on the combined teachings of Linhult, Abrahmsén, and Hober.

c) *Claims 3–8 and 21–26*

Claims 3–8 depend either directly or indirectly from claim 1, and claims 21–26 depend either directly or indirectly from claim 20. These claims further limit the recited processes of claims 1 and 20 whereby the recited matrix retains a percentage of its original binding capacity after certain NaOH exposures.

We are not persuaded by Patent Owner's argument that "none of Petitioners' cited references actually describe a C(G29A)-based SPA ligand, let alone provide alkaline stability data or test results concerning the same, the POSA is simply left to guess at how such a ligand would perform." PO Resp. 50 (citing Ex. 2025 ¶¶ 121–124).

For the reasons discussed above (II.E.5.a), we find that Petitioner has shown by a preponderance of evidence that there is a reason to make the G29A mutation in any one of the SPA IgG binding domains, including domain C, and that the use of the mutated protein would reasonably result in a matrix that can be used to purify IgG. Linhult teaches a CIP protocol with at 0.1 to 0.5M NaOH was a well-known and conventional technique. Pet. 30 (citing Ex. 1004, 1–2, 4–5; Ex. 1002 ¶ 152); Ex. 1004, 6 ("Figure 3, the Z(N23T) mutant shows higher resistance to alkaline conditions than the Z domain when exposed to high pH values."); *see also* Ex. 1006, 39 ("Cleaning-In-Place (CIP) with CIP-buffer with a contact time between column matrix and 0,5 M NaOH of 1 hour"). "Between each cycle [in Linhult], a CIP-step was integrated. The cleaning agent was 0.5 M NaOH and the contact time for each pulse was 30 min." Ex. 1004, 4. "After 16 [CIP] cycles, giving a total exposure time of 7.5 h, the column with the Z(F30A)-matrix shows a 70% decrease in capacity." Ex. 1004, 5; *see also* Ex. 1006, 39 ("Each cycle [in Hober] was repeated 21 times resulting in a total exposure time between the matrix and the sodium hydroxide of 20 hours for each different matrix"). Both Linhult and Hober recognize that repeated exposures of a SPA chromatography ligand leads to a reduction of the binding capacity over time. That Linhult recognizes that additional mutations could further improve alkaline stability does not detract from Linhult's teaching that a composition containing the single G29A mutation in SPA domain B retains IgG binding. Ex. 1004, 6, *see id.* at 4 ("The Z-domain already possesses a significant tolerance to alkaline conditions.").

Petitioner has shown by a preponderance of the evidence of record that there is a reason for making the G29A mutation in *any one* of the four remaining SPA domains in order to produce a SPA product that is more alkaline stable and would reasonably bind IgG. *See* Pet. 26–28, 30; Reply 20–21; *see* Ex. 1002 ¶¶ 115–119.

d) Claims 9, 10, 12–14, 16–19, 27–28, 30–32 and 34–37

Petitioner asserts that Linhult, Abrahmsén, and Hober teaches the additional limitations of the dependent claims. *See* Pet. 26–30, 37–41. In support of these arguments, Petitioner directs our attention to the relevant disclosures of Linhult, Abrahmsén, and Hober and provides a detailed claim analysis addressing how each element of the challenged claims are disclosed by the cited prior art and provides explanation as to why one of ordinary skill in the art would have combined the references to arrive at the claimed subject matter with a reasonable expectation of success. *Id.*

Patent Owner does not offer arguments addressing Petitioner's substantive showing with respect to claims 9, 10, 12–14, 16–19, 27–28, 30–32 and 34–37 separate from its arguments about claims 1, 2, and 20. *See generally* PO Resp.

We have reviewed Petitioner's arguments and the underlying evidence cited in support, which we adopt as our own, and determine that Petitioner establishes that the combination of Linhult, Abrahmsén, and Hober teaches the additional limitations of these dependent claims. *See e.g.*, Pet. 26 (citing Ex. 1004, 4; Ex. 1002 ¶¶ 112–114), *see id.* at 30 (citing Ex. 1002 ¶¶ 128–129); Ex. 1004, 4 ("a multimerization of the domain to achieve a protein A–like molecule"); Ex. 1005, 9:15–10:35. In particular, we note that Hober teaches that monomeric mutant proteins can be combined into multimeric proteins, such as dimers, trimers, tetramers, pentamers, and other multimers. Ex. 1006, 11. Hober also discloses that the multimer comprises mutant monomer units "linked by a stretch of amino acids preferably ranging from 0 to 15 amino acids, such as 5-10." *Id.* Petitioner also provides sufficient explanation as to why one of ordinary skill in the art would have combined the references to arrive at the claimed subject matter with a reasonable expectation of success. Pet. 26–30, 37–41.

Accordingly, we determine that the preponderance of evidence of record supports Petitioner's contentions with respect to claims 9, 10, 12–14, 16–19, 27–28, 30–32 and 34–37.

6. Summary

For the foregoing reasons, we determine that Petitioner has shown by a preponderance of evidence that of claims 1–10, 12–14, 16–28, 30–32, and 34–37 of the '007 patent are unpatentable based on the combination of Linhult, Abrahmsén, and Hober.

For the reasons discussed above, Petitioner has not shown by a preponderance of evidence that of claims 11 and 29 of the '007 patent are unpatentable.

F. *Other Asserted Grounds*

1. The '042 IPR Grounds Based on Other Various Combinations of Linhult, Abrahmsén, and Hober

Petitioner also asserts that claims 1–11 and 20–29 are unpatentable as obvious over Linhult and Abrahmsén (Pet. 18–30); that claims 1–14, 16–32, and 34–37 are unpatentable as obvious over Linhult and Hober (*id.* at 30–48); and that claims 1–14, 16–32, and 34–37 are unpatentable as obvious over Abrahmsén and Hober (*id.* at 49–60) under 35 U.S.C. §103(a).

Because Petitioner has already shown that challenged claims 1–10, 12–14, 16–28, 30–32, and 34–37 are unpatentable over Linhult, Abrahmsén, and Hober as obvious, as discussed supra, we do not reach these claims in these additional asserted grounds as to those claims. *See Beloit Corp. v.*

Valmet Oy, 742 F.2d 1421, 1423 (Fed. Cir. 1984) ("The Commission . . . is at perfect liberty to reach a 'no violation' determination on a single dispositive issue."); *Boston Sci. Scimed, Inc. v. Cook Grp., Inc.*, 809 F. App'x 984, 990 (Fed. Cir. 2020) (recognizing that "[t]he Board has the discretion to decline to decide additional instituted grounds once the petitioner has prevailed on all its challenged claims").

Petitioner has not shown that the challenged claims 11 and 29 are unpatentable over Linhult, Abrahmsén, and Hober. We note that each of Linhult, Abrahmsén, and Hober is silent with respect to Fab biding to a mutant SPA domain. *See above* II.E.5.b. Patent Owner asserts, and we agree, that there is no reasonable expectation that a G29A SPA domain mutant would bind Fab. Specifically, Patent Owner's cited references show that a G29A mutation in domain B results in a domain Z matrix composition that *does not* bind Fab. *See* Ex. 2029, 6 (Fig. 3 (*compare* Panel B-domain B, *with* Panel B-domain Z); Ex. 2013, 3 ("[t]he site responsible for Fab binding is structurally separate from the domain surface that mediates Fcɣ binding"). For the reason discussed above (II.E.5.b), Petitioner has not shown by a preponderance of the evidence of record that a G29A SPA domain mutant would bind a Fab fragment so that a process of using the G29A SPA domain ligand would reasonably result in the purification of the Fab target. That deficiency in Petitioner's showing persists whether the grounds are based on the combination of Linhult, Abrahmsen, and Hober (discussed above) or the related combinations of Linhult combined with Abrahmsen or Hober or Abrahmsen combined with Hober.

2. The ’045 IPR Grounds Based on Various Combinations of Berg[13], Linhult, Abrahmsén, and Hober

Petitioner asserts that claims 1–14, 16–18, 20–32, and 34–36 are unpatentable as obvious over Berg[14] and Linhult (’045 IPR Pet. 21–39); claims 4–8, 10, 19, 22–26, 28, and 37 are unpatentable as obvious over Berg, Linhult, and Hober (*id.* at 39–46); that claims 1–3, 9, 10, 12–14, 16–18, 20, 21, 27, 28, 30–32, and 34–36 are unpatentable as obvious over Berg and Abrahmsén (*id.* at 47–54); and that claims 4–8, 10, 11, 19, 22, 26, 28, 29, and 37 are unpatentable as obvious over Berg, Abrahmsén, and Hober (*id.* at 54–57) under 35 U.S.C. §103(a).

Berg relates to a chromatography matrix to which antibody-binding protein ligands are immobilized. Ex. 1018, Abstract. Petitioner relies on a single paragraph in Berg, paragraph 29, for teaching antibody binding ligands including SPA domain C. *See* ’045 IPR Pet. 24–25, 48. The remainder of the Berg reference is directed to the structure of the chromatography matrix. *See generally* Ex. 1018. A review of Berg shows that SPA is mentioned at three locations in the reference. *See* Ex. 1018 ¶¶ 28, 29, and claim 12. Paragraph 29 of Berg suggests using ligands made up of one or more domains A, B, C, D, and E, and preferably domain B and/or domain C. Ex. 1018 ¶ 29. Just like Linhult, Abrahmsén, and Hober, Berg

[13] Berg et al., US 2006/0134805 A1, published June 22, 2006. Ex. 1018.
[14] We recognize that there is a dispute between the parties whether Berg qualifies as a 35 U.S.C. §102(a) date reference or a §102(b) date reference. *See* Pet; PO Resp., Reply, and Sur-reply. Because we do not need to reach these additional asserted grounds based on Berg beyond addressing whether Berg teaches Fab binding in the context of a mutant SPA domain ligand to establish that Berg does not address the deficiency of the Linhult, Abrahmsén, and Hober combination, we, therefore, do not need to address the prior art status of Berg.

also does not say anything about the ability of a mutant SPA domain ligand, including a domain C ligand, to bind Fab.

Patent Owner, however, cites prior art references to establish that at the time the invention was made there was no expectation that a G29A SPA domain mutant would bind Fab. *See above* II.E.5.b; *see* ’045 IPR PO Resp. 43, 54, 55, 58. Specifically, Patent Owner’s cited references showing that a G29A mutation in domain B results in a domain Z matrix composition that does not bind Fab. *See id.*; Ex. 2029, 6 (Fig. 3 (compare Panel B-domain B, with Panel B-domain Z); Ex. 2013, 3 (“[t]he site responsible for Fab binding is structurally separate from the domain surface that mediates Fcɣ binding”).

Berg, therefore, does not address Fab binding in the context of a mutant SPA domain ligand, specifically domain C, nor does Berg explain why one of ordinary skill in the art would have reasonably expected domain C to retain the ability to bind Fab when other SPA domain mutants do not retain this feature. Because Berg does not address the deficiency of Linhult, Abrahmsén, and Hober as identified by Patent Owner and discussed above (II.E.5.b), any combination of Berg in conjunction with Linhult, Abrahmsén, and/or Hober would not address the missing limitation of claims 11 and 29. Therefore, we do not reach these additional asserted grounds based on Berg beyond addressing whether Berg addresses this missing limitation. *See Beloit Corp.,* 742 F.2d at 1423.

III. CONCLUSION[15]

For the foregoing reasons, we determine that Petitioner has demonstrated by a preponderance of the evidence that claims 1–10, 12–14,

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[15] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this

16–28, 30–32, and 34–37 of the '007 patent are unpatentable, and that claims 11 and 29 have not been shown unpatentable on the bases set forth in the following table.

In summary:

IPR2022-00042

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1–11, 20–29	103(a)	Linhult, Abrahmsén[16]		11, 29
1–14, 16–32, 34–37	103(a)	Linhult, Hober[17]		11, 29
1–14, 16–32, 34–37	103(a)	Linhult, Abrahmsén, Hober	1–10, 12–14, 16–28, 30–32, 34–37	11, 29
1–14, 16–32, 34–37	103(a)	Abrahmsén, Hober[18]		11, 29

decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

[16] As explained above (II.F.1), we do not reach claims 1–10 and 20–28 in this '042 IPR ground because the challenge based on the Linhult, Abrahmsén, and Hober ground is dispositive of these claims.

[17] As explained above (II.F.1), we do not reach claims 1–10, 12–14, 16–28, 30–32, and 34–37 in this '042 IPR ground because the challenge based on the Linhult, Abrahmsén, and Hober ground is dispositive of these challenged claims.

[18] As explained above (II.F.1), we do not reach claims 1–10, 12–14, 16–28, 30–32, and 34–37 in this '042 IPR ground because the challenge based on the Linhult, Abrahmsén, and Hober ground is dispositive of all the challenged *claims*.

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
Overall Outcome			1–10, 12–14, 16–28, 30–32, 34–37	11, 29

IPR2022-00045

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
1–14, 16–18, 20–32, 34–36	103(a)	Berg, Linhult[19]		11, 29
4–8, 10, 19, 22–26, 28, 37	103(a)	Berg, Linhult, Hober[20]		
1–3, 9, 10, 12–14, 16–18, 20, 21, 27, 28, 30–32, 34–36	103(a)	Berg, Abrahmsén[21]		

[19] As explained above (II.F.2), we do not reach claims 1–10, 12–14, 16–18, 20–28, 30–32, and 34–36 in this ’045 IPR ground because the challenge based on the Linhult, Abrahmsén, and Hober ground is dispositive of all the challenged claims.
[20] As explained above (II.F.2), we do not reach this ’045 IPR ground because the challenge based on the Linhult, Abrahmsén, and Hober ground is dispositive of these challenged claims.
[21] As explained above (II.F.2), we do not reach this ’045 IPR ground because the challenge based on the Linhult, Abrahmsén, and Hober ground is dispositive of these challenged claims.

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
4–8, 10, 11, 19, 22, 26, 28, 29, 37	103(a)	Berg, Abrahmsén, Hober[22]		11, 29
Overall Outcome			1–10, 12–14, 16–28, 30–32, 34–37	11, 29

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that the preponderance of the evidence of record has shown that claims 1–10, 12–14, 16–28, 30–32, and 34–37 of the ’007 patent are found unpatentable;

ORDERED that the preponderance of the evidence of record has not shown that claims 11 and 29 of the ’007 patent are found unpatentable; and

FURTHER ORDERED because this is a final written decision, the parties to this proceeding seeking judicial review of our Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

[22] As explained above (II.F.2), we do not reach claims 4–8, 10, 19, 22, 26, 28, 37 in this ’045 IPR ground because the challenge based on the Linhult, Abrahmsén, and Hober ground is dispositive of these challenged claims.

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